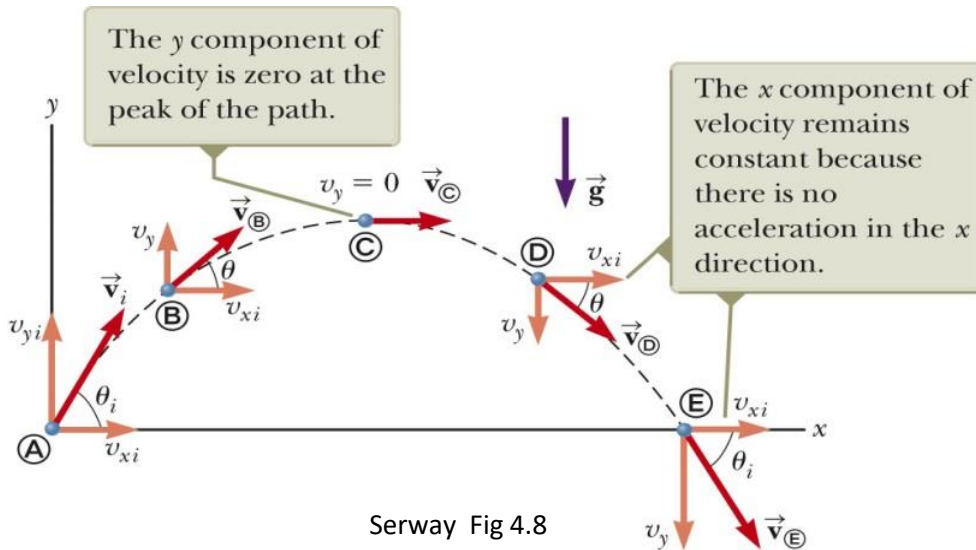


"The secret of success is to know something nobody else knows."

- Aristotle Onassis (1906-1975)

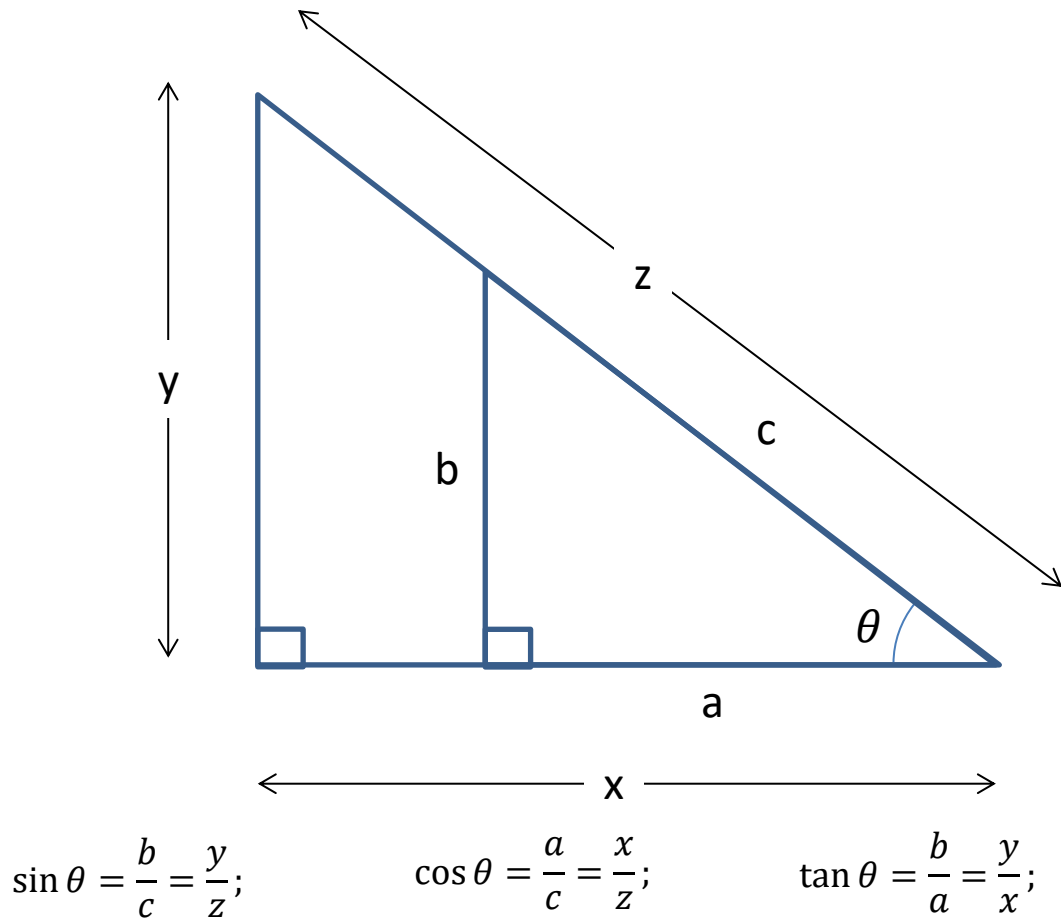


Serway Chap 4

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. apply trigonometry to solve problems in Physics
2. to perform vector addition, vector resolution and vector subtraction to solve problems in Physics
3. use $\hat{i}, \hat{j}, \hat{k}$ notation to analyze problems in Physics
4. analyze projectile motion in 2D

Preparation for 2D systems: Similar Triangles and Trigonometry



Pythagora's theorem

Simple Trigonometric Identities

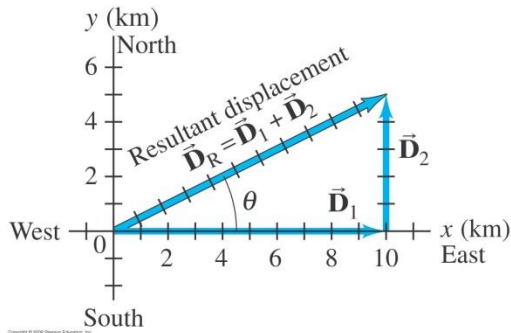
Ratio of Areas

$$\text{If } \frac{y}{b} = \frac{z}{c} = \frac{x}{a} = k, \text{ the ratio } \frac{\text{Area of larger triangle}}{\text{Area of smaller triangle}} = \frac{\frac{1}{2}xy}{\frac{1}{2}ab} =$$

Coordinate Systems and Vectors

A physical quantity with directional properties can be represented by a vector which can depict both the magnitude and direction.

Example: Displacement

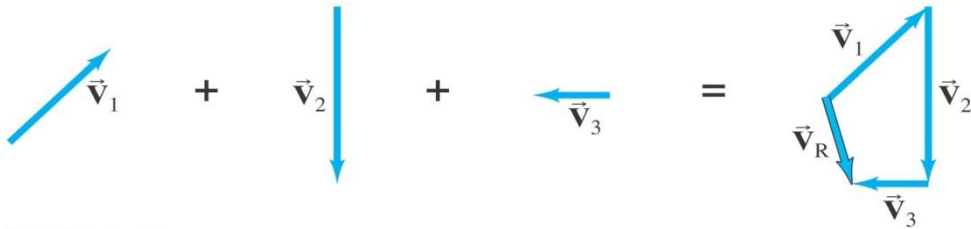


Giancoli Fig 3.3: Displacement

A person walks 10.0 km east and 5.0 km north. The two displacements are represented by vectors $\vec{D}_1$ and $\vec{D}_2$. We can see that the resultant displacement $\vec{D}_R$ has magnitude

and direction

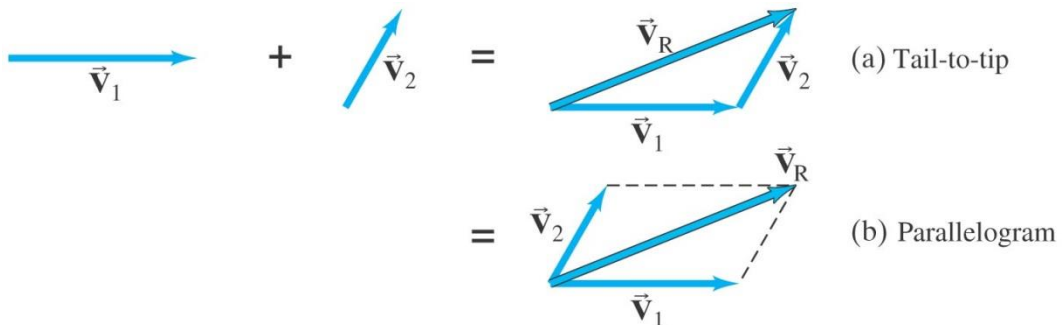
In the absence of reference directions (or coordinate systems), vectors can be combined graphically using the **tail-to-tip method of adding vectors**.



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Giancoli Fig 3.5 / 3.6: Vector Addition

For just two vectors, we can also use the **parallelogram method**.

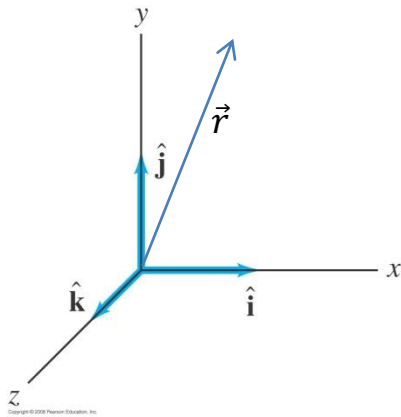


Coordinate Systems and Vectors



Giancoli Fig 3.8: Vector Subtraction

Mutually Perpendicular Coordinate system (3-D)



Giancoli Fig 3.15: 3-D coordinate system

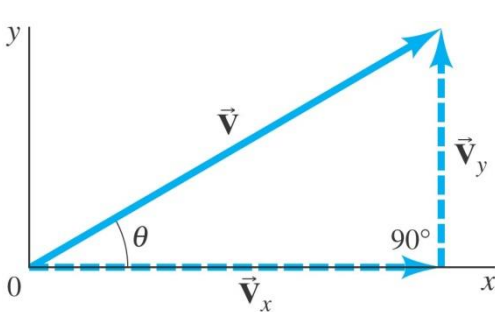
Given a mutually perpendicular coordinate system with basis unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, any vector $\vec{r}$ can be expressed as a column vector of its components.

$$\vec{r} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\vec{r} = a \hat{i} + b \hat{j} + c \hat{k}$$

$$|\vec{r}| = \sqrt{a^2 + b^2 + c^2}$$

Resolution of Vectors in Two Dimensions



$$\sin \theta = \frac{V_y}{V}$$

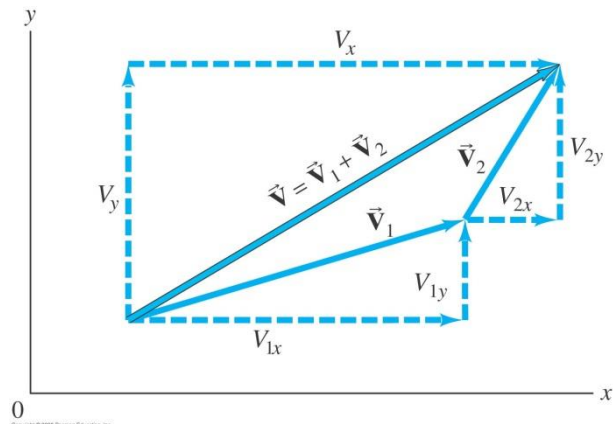
$$\cos \theta = \frac{V_x}{V}$$

$$\tan \theta = \frac{V_y}{V_x}$$

$$V^2 = V_x^2 + V_y^2$$

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Giancoli Fig 3.11: Vector Resolution

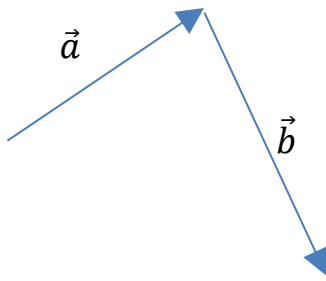


Giancoli Fig 3.12: Vector addition in a coordinate system:

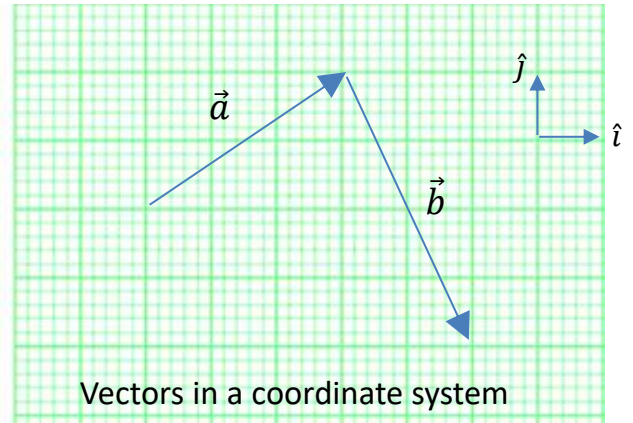
$$\vec{V} = \vec{V}_1 + \vec{V}_2;$$

$$V_x = V_{1x} + V_{2x}; V_y = V_{1y} + V_{2y}$$

Coordinate Systems and Vectors

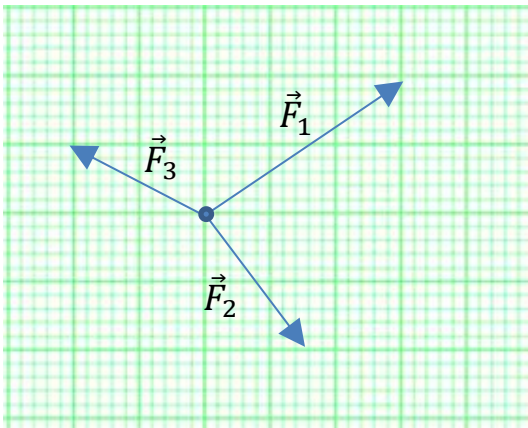


Vectors



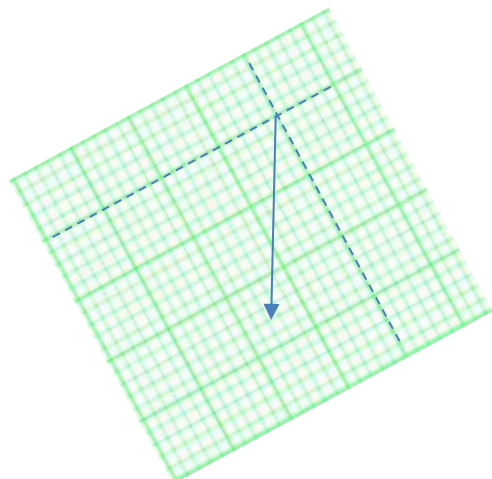
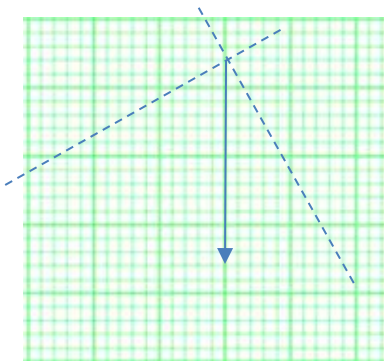
Vectors in a coordinate system

$$\vec{a} + \vec{b} =$$



$$\text{Net } \vec{F} =$$

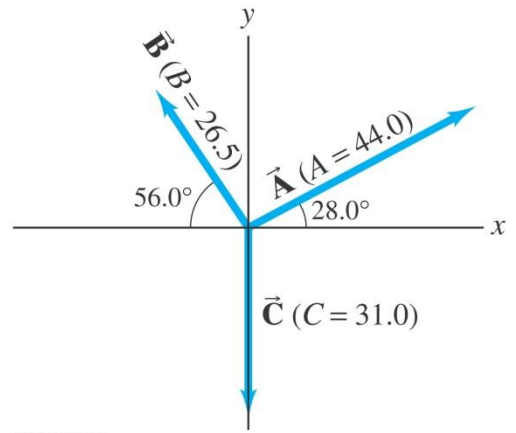
Coordinate system is not unique; we chose the most convenient one to use



Coordinate Systems and Vectors

(Modified) Giancoli Prob 3.10

Three force vectors are shown in the figure. Their magnitudes are given in Newtons. Determine the sum of the three vectors. Give the resultant in terms of (a) components, (b) magnitude and angle with x axis.



Vector Addition and Vector Resolution

Giancoli Example 3-5

The position of a particle as a function of time is given by

$$\vec{r} = [(5.0\text{m/s})t + (6.0\text{ m/s}^2)t^2]\hat{i} + [(7.0\text{m}) - (3.0\text{ m/s}^3)t^3]\hat{j}$$

Where r is in metres and t is in seconds.

- (a) What is the particle's displacement between $t_1 = 2.0\text{ s}$ and $t_2 = 3.0\text{ s}$?
- (b) Determine the particle's instantaneous velocity and acceleration as a function of time.
- (c) Evaluate $\vec{v}$ and $\vec{a}$ at $t = 3.0\text{ s}$.

(a) $\vec{r}(t_1 = 2.0\text{ s}) =$

$\vec{r}(t_2 = 3.0\text{ s}) =$

$\Delta\vec{r} =$

(b) $\vec{v} = \frac{d\vec{r}}{dt} =$

$$\vec{a} = \frac{d\vec{v}}{dt} =$$

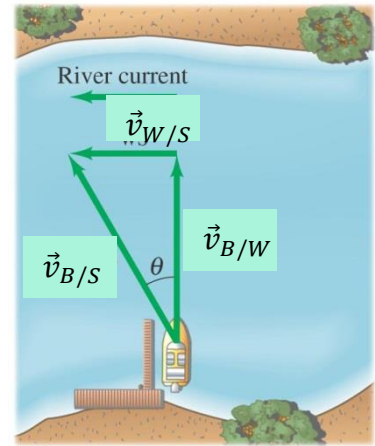
(c) $\vec{v}(t = 3.0\text{ s}) =$

$\vec{a}(t = 3.0\text{ s}) =$

Vector Addition and Vector Resolution

Giancoli Example 3-15 (modified)

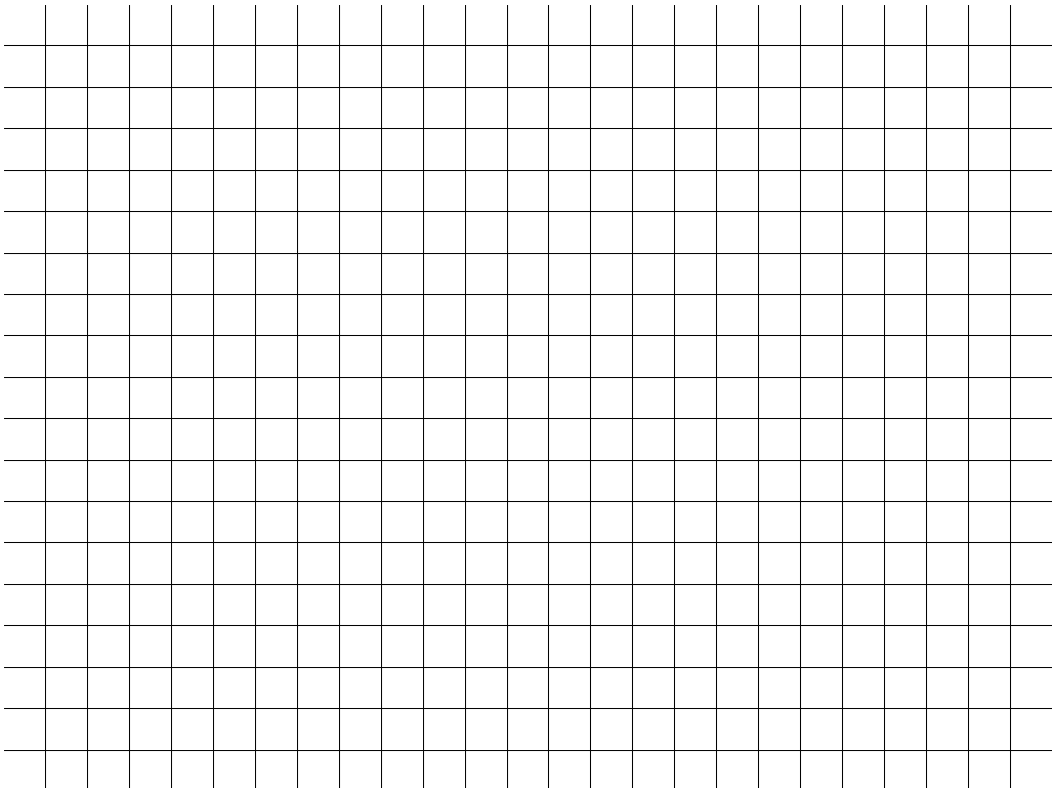
A boat's speed in still water is $v_{B/W} = 2.0 \text{ m/s}$. It heads directly across the river whose current has speed $v_{W/S} = 1.0 \text{ m/s}$. (a) What is the velocity and the magnitude of the boat relative to the shore? (b) If the river is 16 m wide, how long will it take to cross and how far downstream will be the boat?



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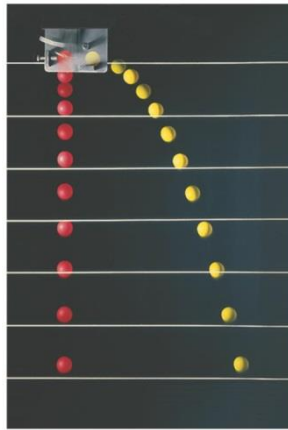
Giancoli Fig 3.34:
Drifting along

Constant velocity in two directions



Motion in Two dimensions (Projectile)

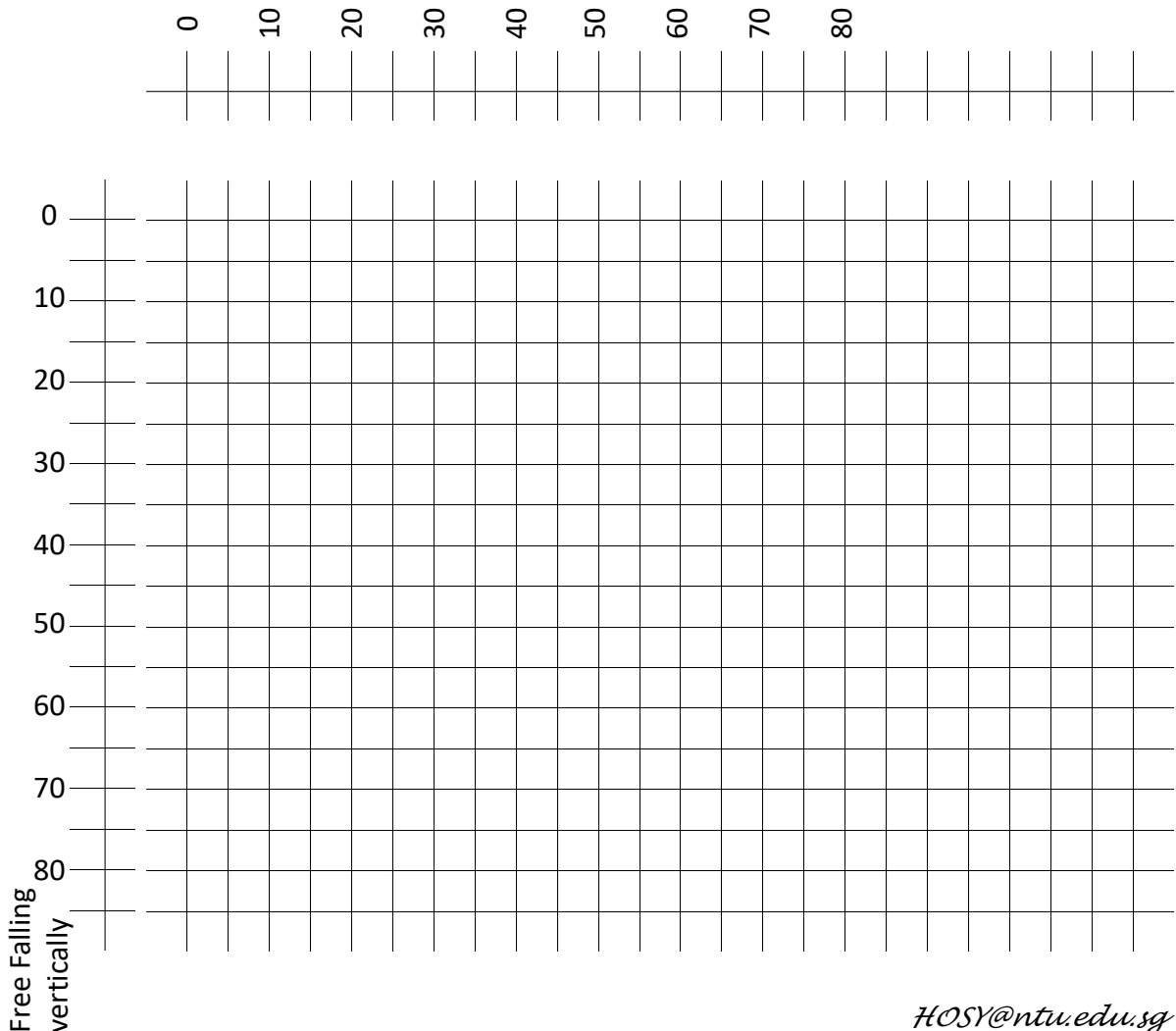
We will analyze the “cannon ball over the cliff” scenario to gain some insights into two dimensional motion.



Giancoli pg 63 Fig 3-21:

Please refer to Giancoli Chap 3.7

Constant horizontal motion

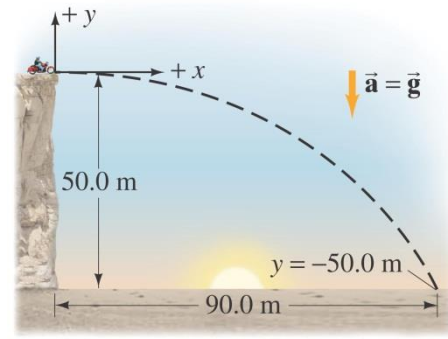


HOSY@ntu.edu.sg

Class Discussion: Motion in Two dimensions (Projectile)

Giancoli Example 3-6

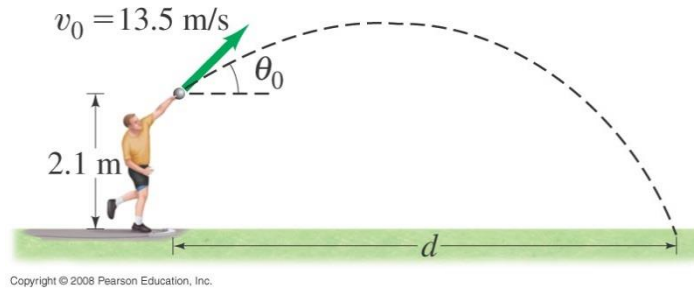
A movie stunt driver on a motorcycle speeds horizontally off a 50.0-m-high cliff. How fast must the motorcycle leave the cliff top to land on level ground below, 90.0 m from the base of the cliff where the cameras are? Ignore air resistance. What is the velocity of the motorcycle just before it hits the the ground below?



Giancoli Fig 3.23:
Projected horizontally over cliff

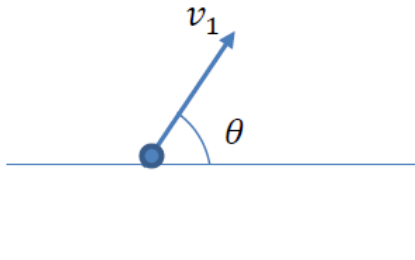
Motion in Two dimensions (Projectile)

[G3.100 modified] A shot-putter throws from a height $h = 2.1\text{ m}$ above the ground as shown in figure, with an initial speed of $v_o = 13.5\text{ m/s}$. If the angle $\theta_o = 30^\circ$, determine the distance d and calculate the velocity of impact (leave your answer in vector form).

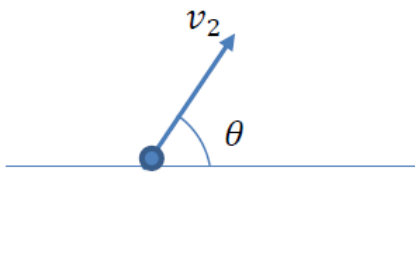


Class Discussion

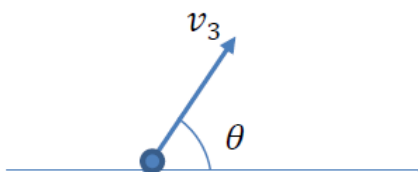
Launching



If $v_1 = 20 \text{ m/s}$ and $\theta = 37^\circ$ and the projectile hits the loop at $y = 3 \text{ m}$, determine the time-of-flight t and the horizontal separation between the launch point and the loop.



If $v_2 = 20 \text{ m/s}$ and $\theta = 37^\circ$ and the projectile hits the loop at the same level of launch, determine the time-of-flight t and the horizontal separation between the launch point and the loop.



If $v_3 = 20 \text{ m/s}$ and $\theta = 37^\circ$ and the projectile hits the loop at $y = -3 \text{ m}$, determine the time-of-flight t and the horizontal separation between the launch point and the loop.



Class Discussion: Motion in Two dimensions (Projectile)



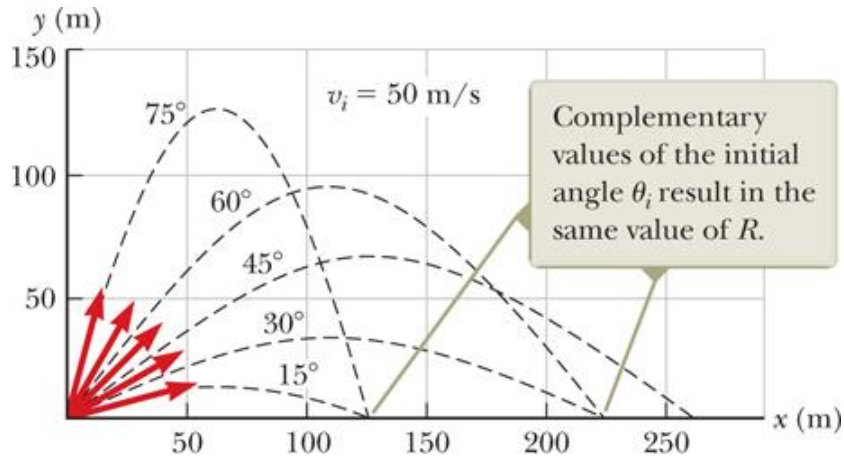
FINAL		DISTANCE
MEN'S SHOT PUT		
1.	KOVACS	
20.90	21.63	21.94 21.95 21.94 22.01
2.	WALSH	22.00m
3.	CROUSER	22.71m
4.	ROMANI	22.53m
5.	HILL	21.65m
6.	BUKOWIECKI	21.46m
7.	GILL	21.45m
8.	ENEKWECHI	21.18m
9.	NEDOW	20.85m
10.	STANĚK	20.79m
11.	MIHALJEVIĆ	20.48m
WCH DOHA 2019		

Based on the information that you can extract from the Youtube video, determine the angle and speed that the athlete release the shot-put. Assume that the shot-put is released from a height of 2.1 m as in the question on the previous page

<https://www.youtube.com/watch?v=SEFOEWCsq6Q> 8:15



Motion in Two dimensions (Projectile)



Serway Fig 4.11
Projectile on a Flat Surface

(a) Derive a formula for the horizontal range R of a projectile in terms of its initial speed v_i and angle θ_i . The horizontal range is defined as the horizontal distance the projectile travels before returning to its original height (which is typically the ground); that is, $y(\text{final}) = y_i$. (b) Suppose one of the cannons had a muzzle speed, v_i , of 50.0 m/s . What is the horizontal range when $\theta_i = 45^\circ$, 75° , 15° ? What is the angle of projection if the target is (i) 200 m away, (ii) 300 m away?

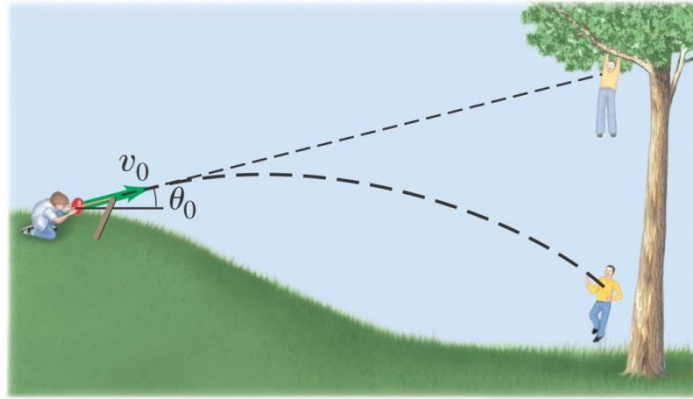
Just for Fun: Motion in Two dimensions (Projectile)

Giancoli Example 3-9; Prob 49

A boy on a small hill aims his water-balloon slingshot at an angle θ_0 above the horizontal, straight at a second boy hanging from a tree branch distance d away. At the moment the water-balloon is released, the second boy lets go and falls from the tree. Will he be hit by the water-balloon?



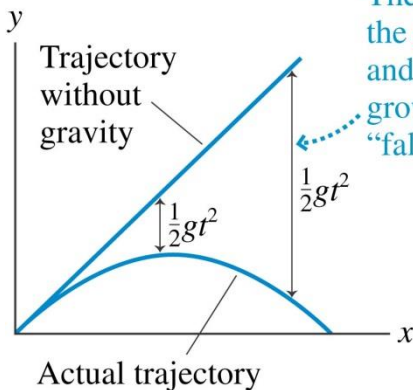
<https://www.youtube.com/watch?v=0jGZnMf3rPo>



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Giancoli Fig 3.26

(a)

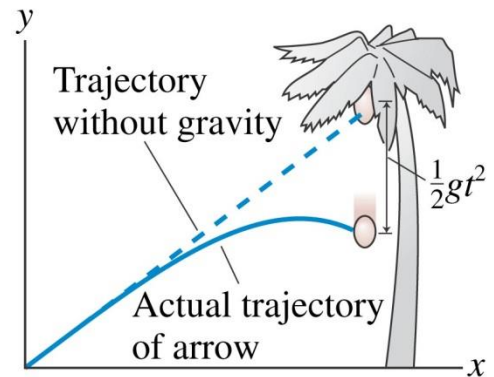


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Knight Fig 4.20

The distance between the gravity-free trajectory and the actual trajectory grows as the particle "falls" $\frac{1}{2}gt^2$.

(b)



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Knight Fig 4.20

2026S1 PH1012: Physics A
Newton's Laws

Dr Ho Shen Yong
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Week 3 & 4
Serway Chap 5

"If I have seen further it is by standing on the shoulders of giants."

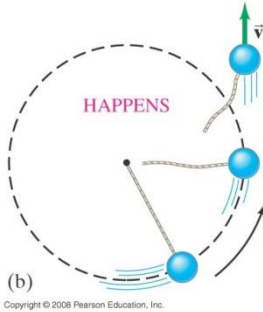
- Isaac Newton (1643 - 1727) in Letter to Robert Hooke (15 February 1676)

"We first make our habits, then our habits make us."

- John Dryden (1631 – 1700)

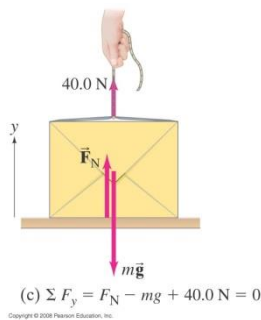
Outline

Newton's three laws of motion... In particular, $\sum \vec{F} = m\vec{a}$

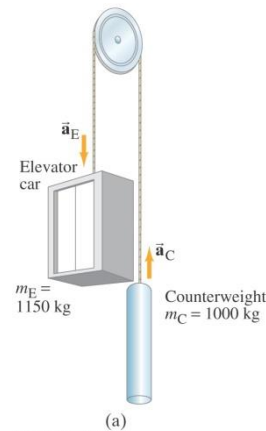


Top view

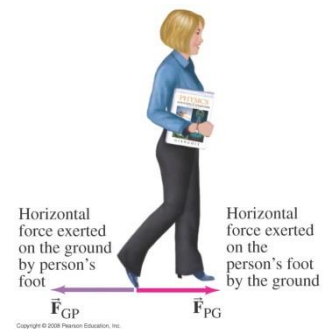
Giancoli Fig 5.16b



Giancoli Fig 4.15



Giancoli Fig 4.23



$$\vec{F}_{GP} = -\vec{F}_{PG}$$

Giancoli Fig 4.11

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. identify forces (e.g. weight, tension, normal force, friction) acting on a object in various context
2. apply Newton's first law of motion to make predictions of the motion of a given object
3. apply Newton's second law of motion to analyze the forces and acceleration acting on an object or system of objects
4. apply Newton's third law of motion to analyze the forces acting on a system of objects
5. apply Newton's laws of motion whenever appropriate to analyze forces and motion of objects
6. analyze forces acting on an object on an incline plane and objects connected by pulley(s)

Forces (Knight 4.1)

- A **force** is a *push* or a *pull*.



- A force acts on an object.

- Every force has an **agent**, something that acts or pushes or pulls.



- A **force** is a *vector*. The general symbol for a force is the vector symbol $\vec{F}$. The size or strength of such a force is its magnitude F .
- **Contact forces** are forces that act on an object by touching it at a point of contact.
- **Long-range forces** are forces that act on an object without physical contact.



Contact Forces



Long Range Forces

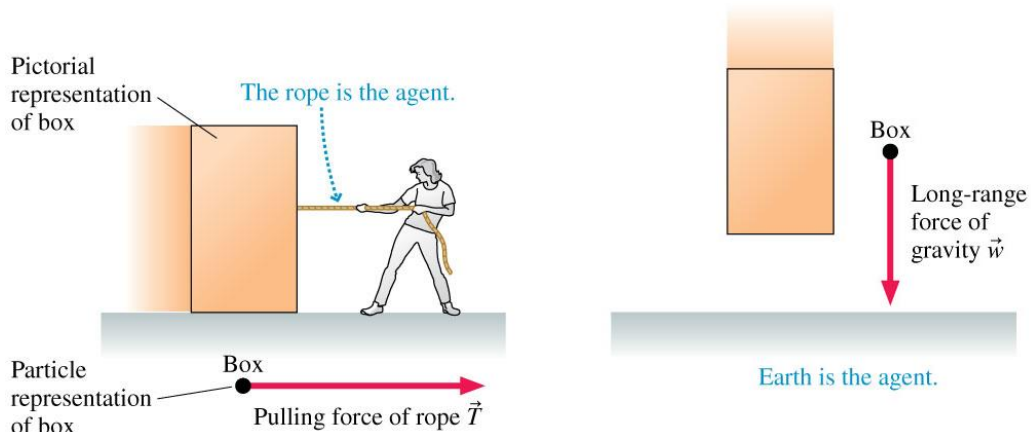
Forces (Knight 4.1)

TACTICS BOX 4.1 Drawing force vectors

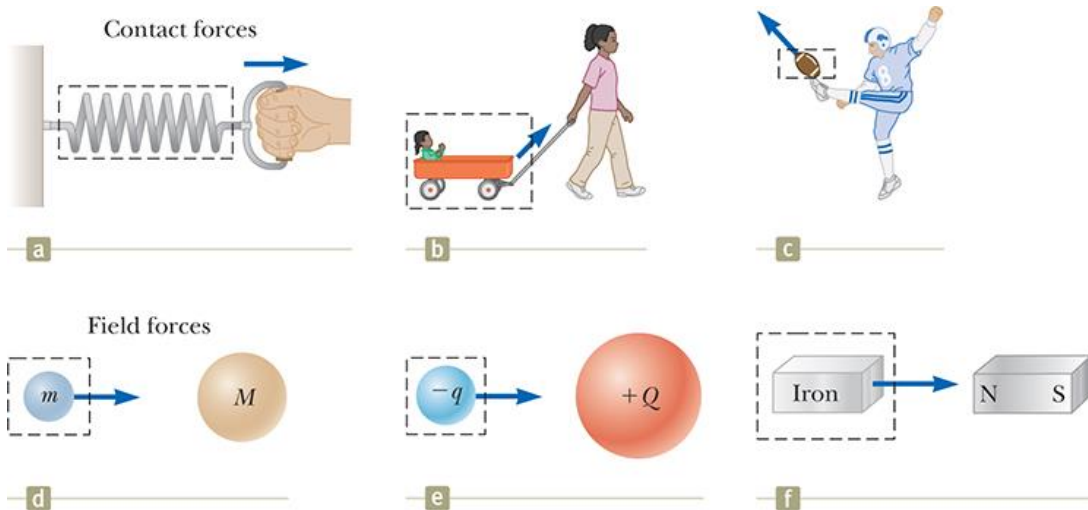


- 1 Represent the object as a particle.
- 2 Place the *tail* of the force vector on the particle.
- 3 Draw the force vector as an arrow pointing in the direction that the force acts, and with a length proportional to the size of the force.
- 4 Give the vector an appropriate label.

Exercise 1



Knight Fig 4.3

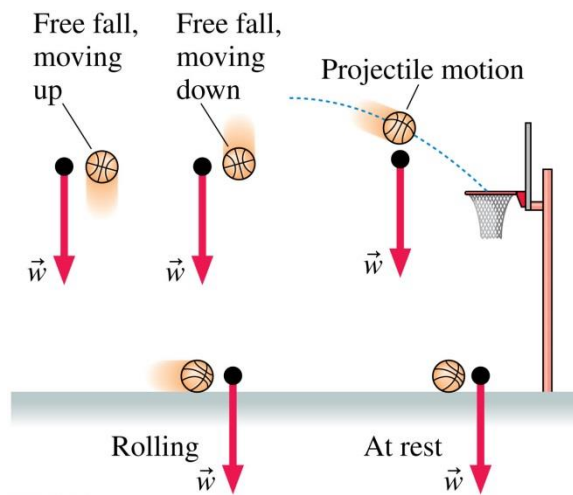


Serway Fig 5.1

Types of Forces (Knight)

Weight

- The gravitational pull of the earth on an object on or near the surface of the earth is called **weight**.
- The agent for the weight forces is the *entire earth* pulling on an object.
- **An object's weight vector always points vertically downward**, no matter how the object is moving.



Knight Fig 4.5

Upthrust / Bouyant Force

- **Archimedes' Principle:** Any object, wholly or partially immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object.
1. Volume of object submerged in the fluid = Volume of fluid displaced;
 2. Knowing the displaced volume $V_{\text{submerged}}$ and density ρ_{fluid} of the fluid, we can compute the weight of fluid which gives the upthrust:

$$\text{Upthrust} = (\rho_{\text{fluid}} V_{\text{submerged}}) g$$



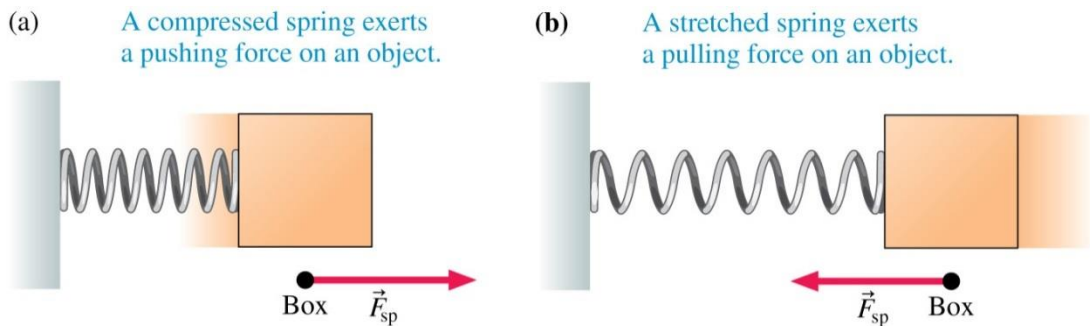
More Forces (Knight)

Spring Force

- Springs come in in many forms. When deflected, they push or pull with a spring force.
- The magnitude of the force is dependent on the extension or compression of the spring.
- If the magnitude of the spring force F_{sp} is directly proportional to extension / compression x , we say that the spring obeys Hooke's law:

$$|F_{sp}| = kx$$

where k is known as the spring constant – the stiffer the spring, the larger will be the value of k .

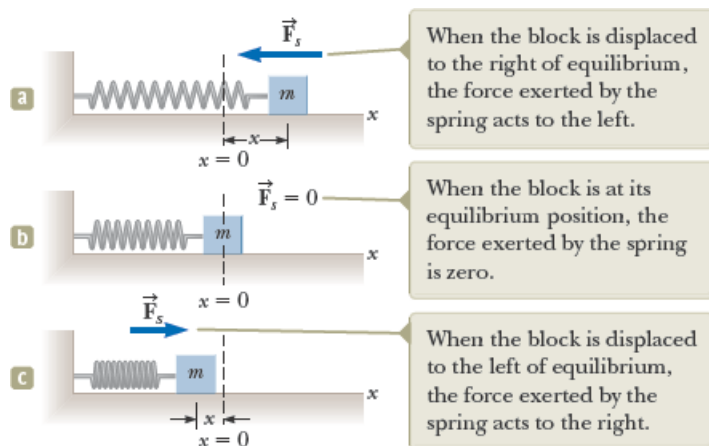


Knight Fig 4.6

Sometimes, we write the vector version of the equation:

$$\vec{F}_{sp} = -k\vec{x}$$

The negative sign :

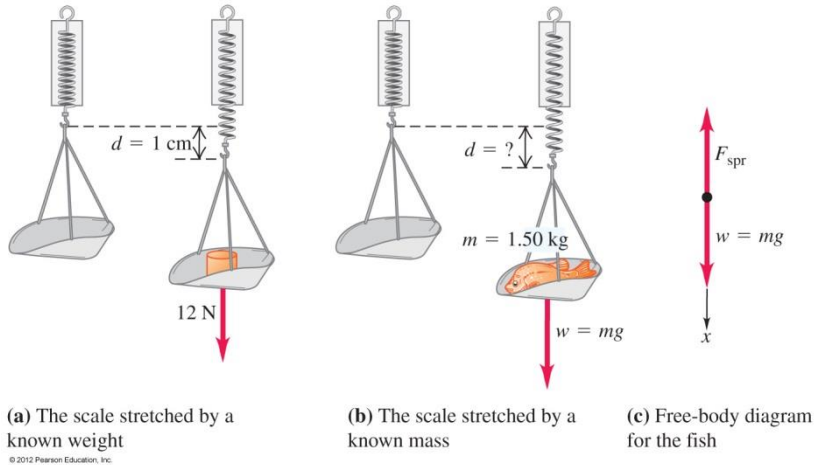


Serway Fig 15.1

Spring force

Example

A spring balance used to weigh fish is built with a spring that stretches 1.00 cm when a 12.0 N weight is placed in the pan. When the 12.0 N weight is placed with a 1.50 kg fish, what distance d does the spring stretch?



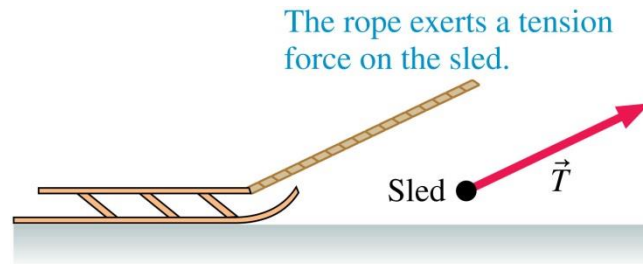
Giancoli Example 14-1: Car springs.

When a family of four with a total mass of 200 kg step into their 1200-kg car, the car's springs compress 3.0 cm. (a) What is the spring constant of the car's springs, assuming they act as a single spring? (b) How far will the car lower if loaded with 300 kg rather than 200 kg?



Tension Force

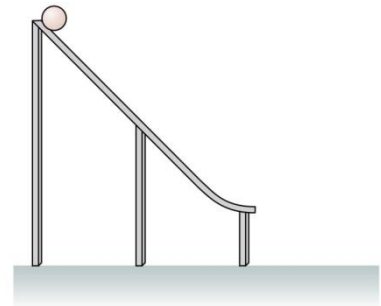
- When a string or rope or wire pulls on an object, it exerts a contact force that we call the **tension force**.
- **The direction of the tension force is always in the direction of the string or rope.**
- The direction of the tension is always pointing away from the object.



Knight Fig 4.7

Exercise

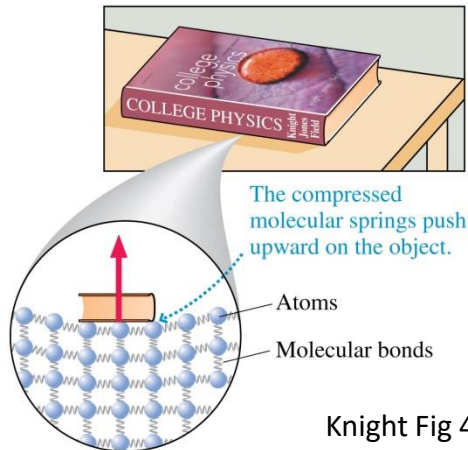
A ball rolls down an incline and off a horizontal ramp. Ignoring air resistance, what force or forces act on the ball as it moves through the air just after leaving the horizontal ramp?



- A. The weight of the ball acting vertically down.
- B. A horizontal force that maintains the motion.
- C. A force whose direction changes as the direction of motion changes.
- D. The weight of the ball and a horizontal force.
- E. The weight of the ball and a force in the direction of motion.

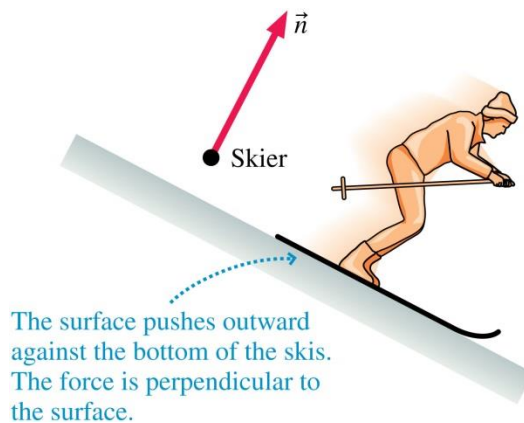
Normal Force

- The force exerted on an object that is pressing against a surface is in a direction *perpendicular* to the surface.
- The **normal force** is the force exerted by a surface (the agent) against an object that is pressing against the surface.



Knight Fig 4.9

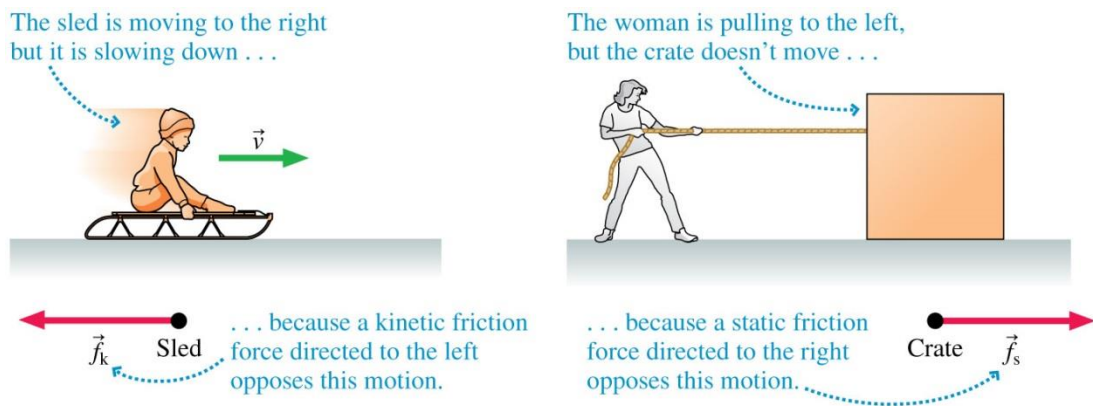
- The normal force is responsible for the “solidness” of solids.
- The symbol for the normal force is $\vec{F}_N$ or $\vec{n}$.



Knight Fig 4.11

Frictional Force

- Friction, like the normal force, is exerted by a surface.
- The frictional force is always parallel to the surface.
- *Kinetic friction*, denoted by $\vec{f}_k$, acts as an object slides across a surface. Kinetic friction is a force that always “opposes the motion.”
- *Static friction*, denoted by $\vec{f}_s$, is the force that keeps an object “stuck” on a surface and prevents its motion relative to the surface. Static friction points in the direction necessary to *prevent* motion.



Knight Fig 4.12

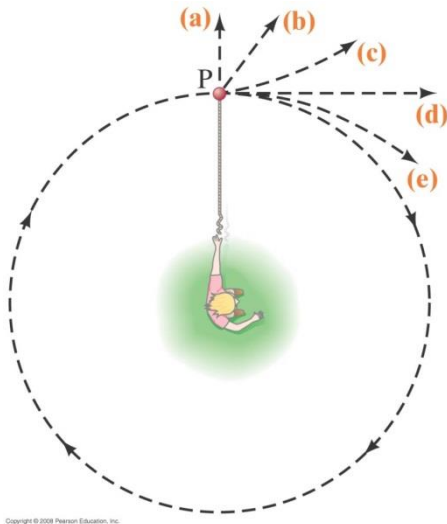
$$\text{Friction} = \mu \times \text{Normal Force}$$

- Here, μ is the coefficient of friction – it is dependent on the nature of the two surfaces touching each other.
- Note that the magnitude of frictional force is INDEPENDENT of area of contact and velocity.

If the coefficient of static friction between a 22kg crate and the floor is 0.30, what is the maximum static friction force between the crate and the floor?

Newton's First Law of Motion

First Law (Inertia): An object at rest remains at rest and an object in motion will continue in motion with a constant velocity (i.e., constant speed in a straight line) unless it experiences a non-zero net external force.



Giancoli Chap 5

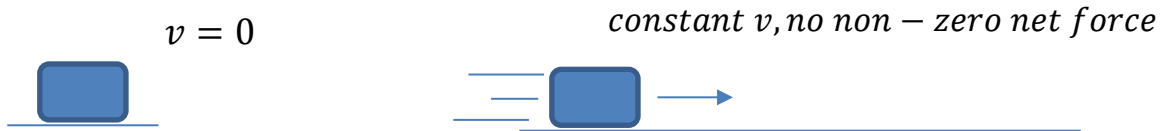


<https://www.youtube.com/watch?v=bBvaHLqmpbE>

SMRT bus 700A flipped along Dairy Farm Road caught on video - 24Jul2013

1. Newton's First Law:

First Law (Inertia): An object at rest remains at rest and an object in motion will continue in motion with a constant velocity (i.e., constant speed in a straight line) unless it experiences a non-zero net external force.



Remarks:

1. The first law involves a property of objects known as inertia - its tendency to resist any change in its state of motion. The physical quantity associated with inertia is mass measured in SI units kilogram (kg). It is an intrinsic property of the object that does not vary with location, surroundings etc.
2. (Constant velocity) If either the direction or speed of an object is not constant, we can infer that there is a non-zero net external force acting on it.
3. (External force) Forces acting on the object by other objects.
4. (Non-zero net) There can be several different external forces acting on an object but if the vectorial sum of these forces is zero, there will be no change in the state of motion of the object.
5. Conversely, an object moving at constant velocity has no net external force acting on it.
6. "Reference frame" used for ascertaining the motion of the object must not be accelerating.

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Some daily occurrences explained using Newton's First Law

1. The 'being pushed forward' sensation of the passengers when a moving car brakes.
2. Pen dropped on an airplane flying at constant velocity doesn't fly to back of the plane.
3. Another way to get the last bit of the (stubborn) ketchup / shampoo out of a bottle.

Example: Free Falling under gravity.



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Example: Cyclist racing downhill crashes into moving car



<https://www.youtube.com/watch?v=gO0i6hjJXlk>



Cyclist Overruns Corner and Crashes Into Car ...and then two more hit motorbike!



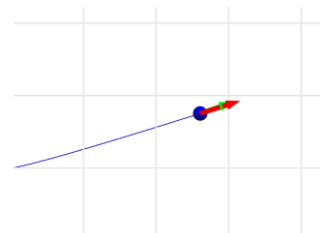
Cyclist Overruns Corner and Crashes Into Car ...and then two more hit motorbike!

Save the Cyclist:

<https://www.glowscript.org/#/user/matterandinteractions/folder/matterandinteractions/program/02-Newton-grid>

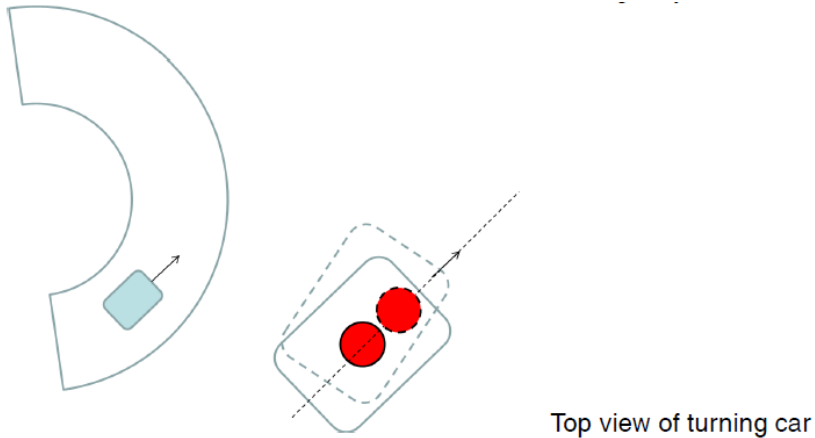


Using the applications in the above link, explore the ways that the crash can be avoided.

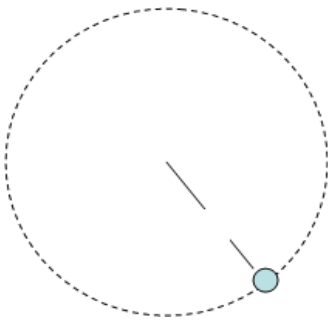


1. Newton's First Law: Applications in calculations

1. When a car turns round a bend, which way do the passengers tend to lean?



2. A ball is swirled in a vertical circle but the string broke at shown below. If you want to analyse the subsequent motion, what is the initial velocity that you should use?



Newton's First Law of Motion

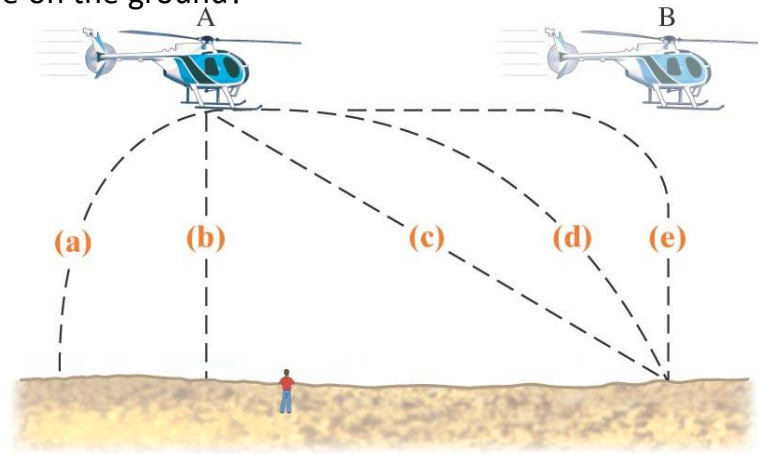
A bowling ball dropped out of a helicopter that is flying at constant velocity v . Which of the following best shows the path of the bowling ball as observed by someone on the ground?



https://www.youtube.com/watch?v=7_Aj9Jw5MXo&t=265s

4min 29s

Giancoli Chap 3



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$t = 0 \text{ units}$

$t = 1 \text{ unit}$

$t = 2 \text{ units}$

$t = 3 \text{ units}$

$t = 4 \text{ units}$



Pearl Harbor Attack R1/2 250168-04 | Footage Farm

<https://www.youtube.com/watch?v=Q9dEnbBMyF0&t=560s>

9 min 23 s

What is Momentum?

Momentum $\vec{p}$ of an object is given by the product of its mass m with its velocity $\vec{v}$:

$$\vec{p} = m\vec{v}$$

Remarks:

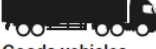
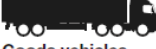
1. [Intuitively] If you have a choice of stopping a mouse or an elephant running towards you at the same velocity [say 10 m/s], which would you choose?

2. Units for momentum: kg m/s.



<https://www.youtube.com/watch?v=6dOYAeRb13E>
College Football Highlights 2014-15 | Pump Up (HD)

Speed Limits

	Built-up areas*	Single carriage-ways	Dual carriage-ways	Motorways
Type of vehicle	mph (km/h)	mph (km/h)	mph (km/h)	mph (km/h)
 Cars & motorcycles (including car-derived vans up to 2 tonnes maximum laden weight)	30 (48)	60 (96)	70 (112)	70 (112)
 Cars towing caravans or trailers (including car-derived vans and motorcycles)	30 (48)	50 (80)	60 (96)	60 (96)
 Buses, coaches and minibuses (not exceeding 12 metres in overall length)	30 (48)	50 (80)	60 (96)	70 (112)
 Goods vehicles (not exceeding 7.5 tonnes maximum laden weight)	30 (48)	50 (80)	60 (96)	70[†] (112)
 Goods vehicles (exceeding 7.5 tonnes maximum laden weight) in England and Wales	30 (48)	50 (80)	60 (96)	60 (96)
 Goods vehicles (exceeding 7.5 tonnes maximum laden weight) in Scotland	30 (48)	40 (64)	50 (80)	60 (96)

* The 30 mph limit usually applies to all traffic on all roads with street lighting unless signs show otherwise.
† 60 mph (96 km/h) if articulated or towing a trailer.



<https://www.youtube.com/watch?v=MmQbEgUVz40>

2. Newton's Second Law

The rate of change of linear momentum of an object is proportional to the net force (vector sum of all forces) acting on it and occurs in the direction of the net force:

$$F = \frac{d(m\vec{v})}{dt}.$$

If the mass is constant, we can write

$$F = \frac{m\vec{v}_f - m\vec{v}_i}{\Delta t} = m \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = m\vec{a}.$$

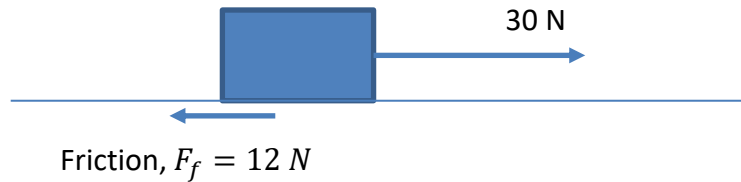
Here, the forces are in units of Newtons (N), the mass is in kilograms (kg), velocity in metres per second (m/s) and acceleration in (m/s^2). We will see a more complete version later (with varying mass).

Example:

Calculate the momentum of a 1200 kg car moving at 72 km/h on the expressway. What is the momentum of the car when it is at rest? If the car stops in 10 s, what is average force exerting on it? What if it stops in 3 s?

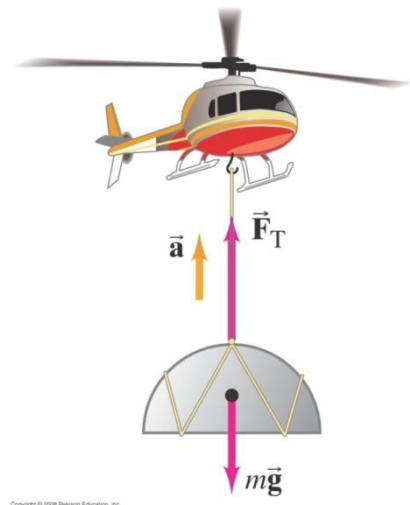
Newton's Second Law of Motion

Example:



A horizontal force of 30 N acts on a block of mass 5 kg in the +ve x direction. A coefficient of frictional force of 12 N acts in the opposite direction. Determine the acceleration of the block.

Example: [G4.78 modified] A 7650-kg helicopter accelerates upward at 0.80 m/s^2 while lifting a 1250-kg frame at a construction site. What is the tension in the cable (ignore its mass) that connects the frame to the helicopter?



Giancoli Fig 4.59



<https://www.youtube.com/watch?v=mzSqLCBP0Qk&t=605s>

See 10 min 5s

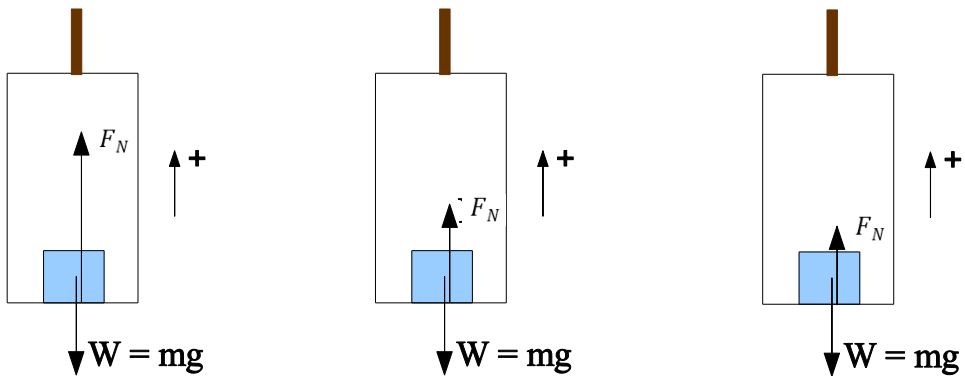
Forces: What does a weighing scale always read?



1. My weight, mg .
2. How hard I press down on the scale, Normal force (F_N).
3. The net force of weight and normal force.
4. Don't know.

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2.1 Forces: Another application - Object in Elevator 1



$$F_{net} = F_N - W$$

Applying Newton's second law:

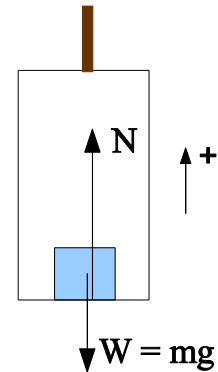
$$F_{net} = F_N - W = ma$$

Forces: Another application - Object in Elevator 2

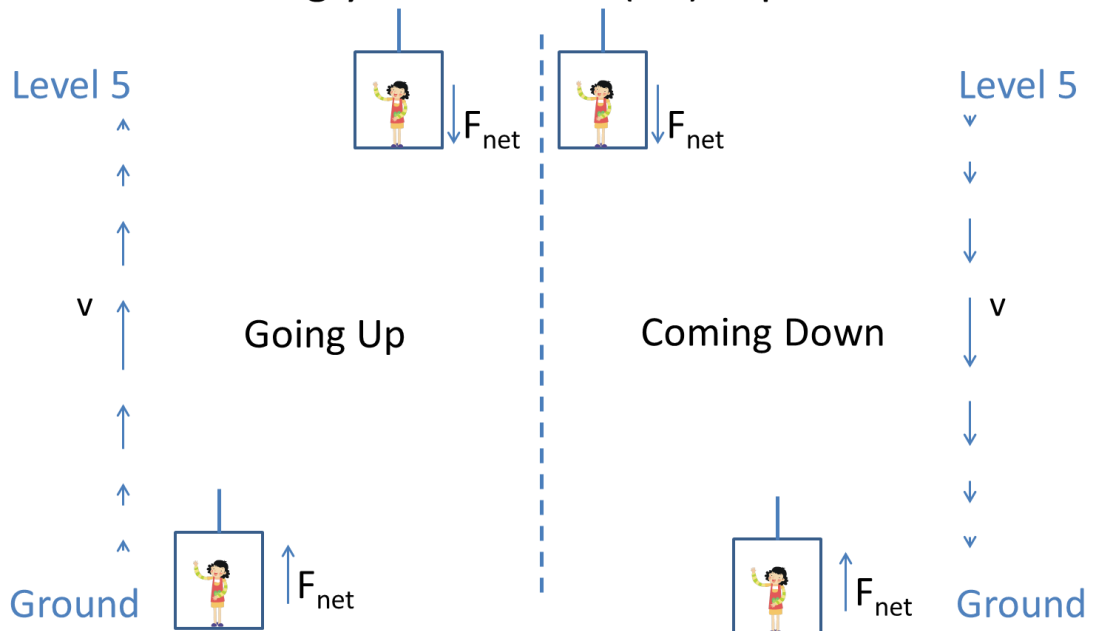
Consider this case where $F_N - W > 0$ and hence $a > 0$:

Is the elevator speeding up or slowing down?

- (A) Speeding up
- (B) Slowing down
- (C) Not enough information
- (D) No idea at all



Recalling your elevator (lift) experience.



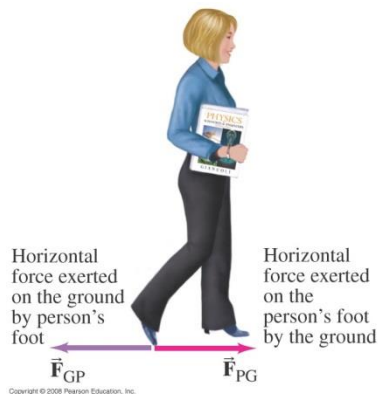
Newton's third law of motion

Third Law: If object A exerts a force on object B (an "action"), then object B exerts a force on object A (a "reaction"). These two forces are equal in magnitude but opposite in direction.

Remarks:

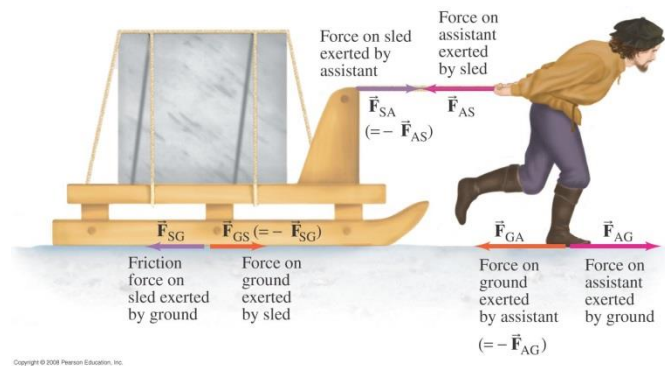
1. The two forces are always opposite in direction.
2. The two forces are at all times equal in magnitude.
3. The forces must act on different bodies.
4. The forces must be of the same type.

Newton's third law of motion



$$\vec{F}_{GP} = -\vec{F}_{PG}$$

Giancoli Fig 4.11



Giancoli Fig 4.12

Application Newton's 3rd law: Analyzing 'Apple on Table'

An apple is at rest on a table. We note that its weight mg and the normal contact force F_N are equal and pointing in opposite direction.

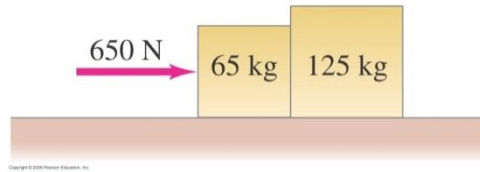
Does the weight mg and normal force F_N form an action-reaction pair?



My finger exerts a force of 1 N on a block. According to Newton's 3rd law, there is a force of 1N in the opposite direction. So why does the block accelerate?

Two objects of Interest - Pushing two blocks

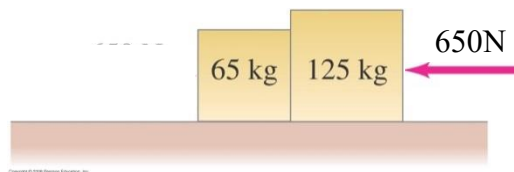
(Modified) Giancoli Prob 5.17 Two crates, of mass 65 kg and 125 kg, are in contact and at rest on a horizontal frictionless surface. A 650-N force is exerted on the 65-kg crate. Calculate (a) the acceleration of the system, and (b) the force that each crate exerts on the other. (c) [Class Discussion] Repeat with the crates reversed.



Considering only horizontal forces:

Equation for 65 kg crate:

Equation for 125 kg crate:



Two objects of Interest – Two pails hanging

(Modified) Giancoli Prob 4.33

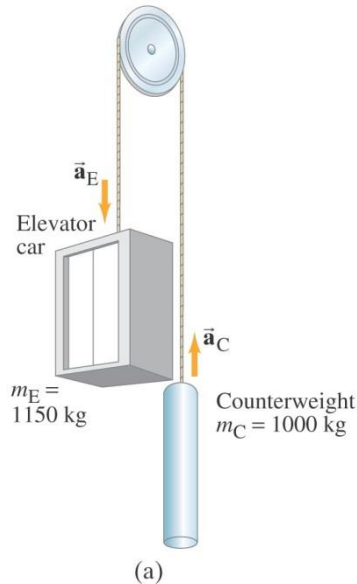
One 3.2-kg paint (bottom) bucket is hanging by a massless cord from another 4.8-kg paint bucket, also hanging by a massless cord, as shown in the figure. (a) If the buckets are at rest, what is the tension in each cord? (b) If the two buckets are pulled upward with an acceleration of 1.25 m s^{-2} by the upper cord, calculate the tension in each cord. (c) Without doing any calculation, deduce how will tensions be if the buckets are accelerating at 1.25 m s^{-2} downwards instead.



Giancoli Fig 4.37: Two pails

Giancoli: Example 4-13: Elevator and counterweight (Atwood's machine).

A system of two objects suspended over a pulley by a flexible cable is sometimes referred to as an Atwood's machine. Here, let the mass of the counterweight be 1000 kg. Assume the mass of the empty elevator is 850 kg, and its mass when carrying four passengers is 1150 kg. For the latter case calculate (a) the acceleration of the elevator and (b) the tension in the cable.

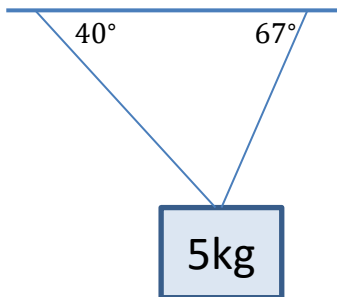
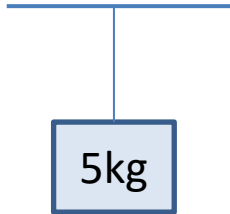


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Giancoli pg 99 Fig 4.23

System of Forces (Two-dimensional Systems)

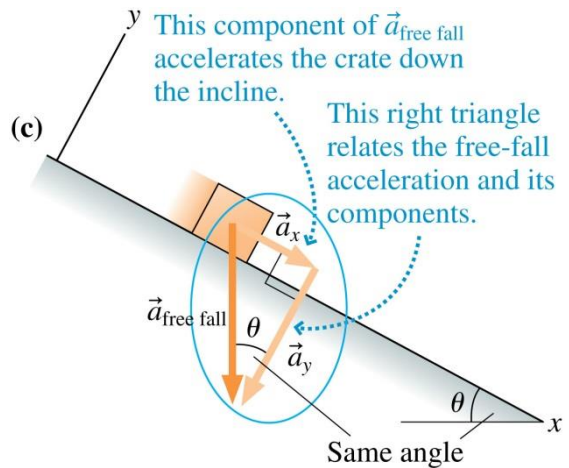
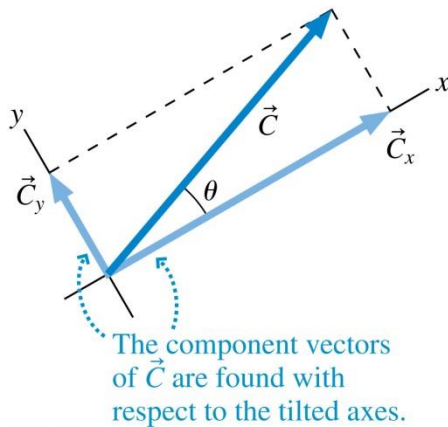
Find the tension in each of the strings in the following systems



Vector Addition and Vector Resolution (Two-dimensional Systems)

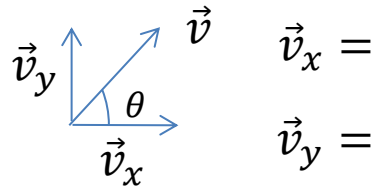
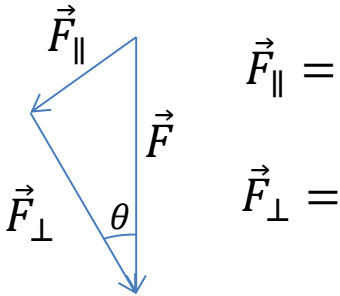
Tilted Axes (Knight)

- For motion on a slope, it is often most convenient to put the x-axis along the slope.
- When we add the y-axis, this gives us a tilted coordinate system.
- Finding components with tilted axes is done the same way as with horizontal and vertical axes. The components are parallel to the tilted axes and the angles are measured from the tilted axes.

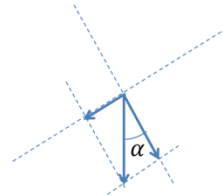
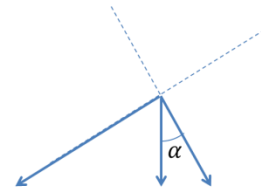
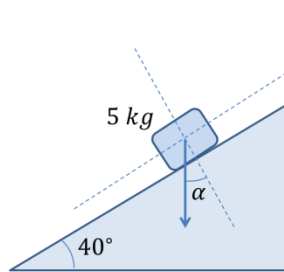
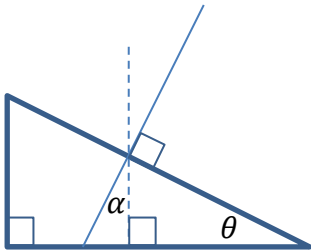


Geometry and Trigonometry / Resolution of Forces

Write down the components of the vectors:



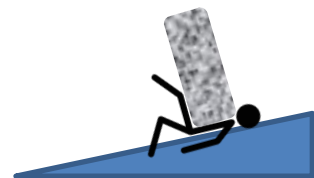
If $\theta = 27^\circ$, what is the angle α ?



In which case does the man require greater effort to prevent the block from sliding down?

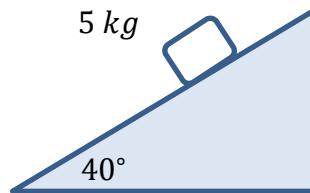


In which case is the man experiencing more 'squashed' by the block?



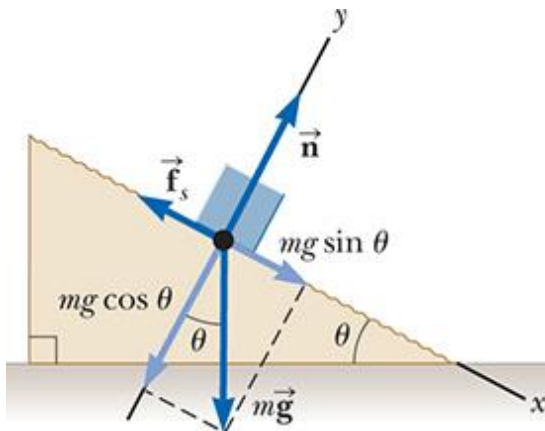
System of Forces (Two-dimensional Systems)

The frictional forces in each of the cases below can be neglected. Compute the acceleration of the blocks.

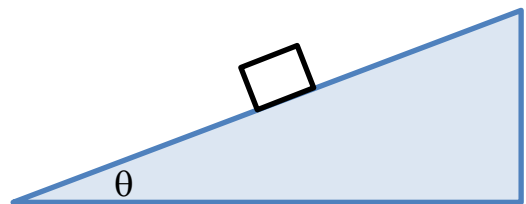


Dealing with friction on an inclined plane

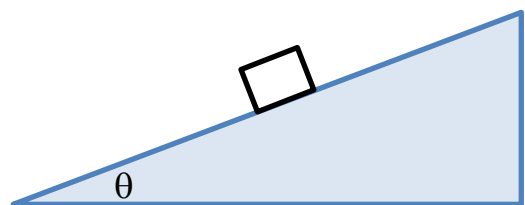
As we just learnt, the frictional force is dependent on the normal force F_N . As the angle θ of the slope is varied, the normal force (hence, frictional force) will change. We will analyse explicitly how the (static/kinetic) frictional force depends on θ .



Serway 5.19



On the verge of sliding down

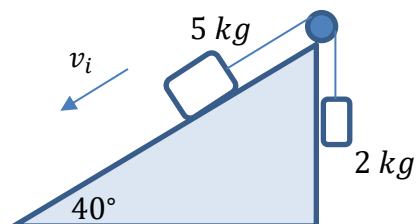


Already sliding down

Dealing with friction on an inclined plane

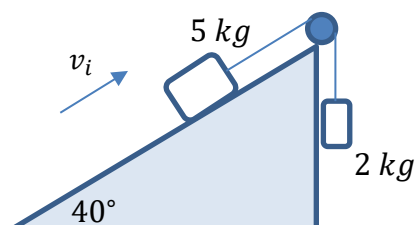
If the coefficient of kinetic friction between the 5 kg block and the slope is 0.2, now determine the acceleration of the blocks. The 5 kg block is initially sliding down slope in this case.

Equation for 5 kg block:



Equation for 2 kg block:

[Class Discussion] In another case, the 5 kg block is initially sliding up slope. Determine the acceleration and describe the subsequent motion of the system.



Friction

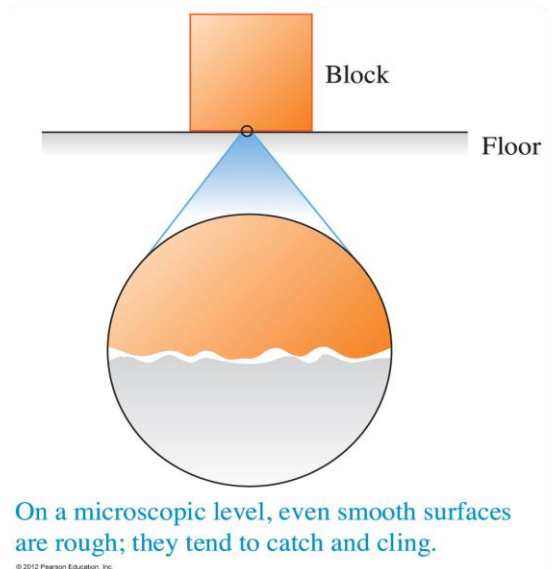
Some aspects of frictional forces

The science of friction is called **tribology**. We make some general observations (not principles) based on experimental results:

1. It is a force that arises due to contact between two surfaces and will vary with combinations of surfaces. For example, frictional forces between wood and wood is different from frictional forces between steel and wood.
2. The magnitude of the frictional force is proportional to the magnitude of the normal force.
3. Frictional force is independent of the size of the contact area between the object and the surface.
4. Frictional force is independent of velocity.

Microscopic Origin of Friction Forces

1. Actual area of contact is much less than true area of surface.
2. Microscopically, contact points undergo great stress resulting in plastic deformation - contact points become 'cold-welded' together due to strong intermolecular forces between the molecules from the two surfaces.
3. When one body (say a metal) is pulled across another, frictional resistance is associated with rupturing of these thousands of tiny welds and they reform as new contacts are made.
4. This 'catch' and 'cling' events are responsible for the heat generation, and sound when dry surfaces make when sliding across each other.

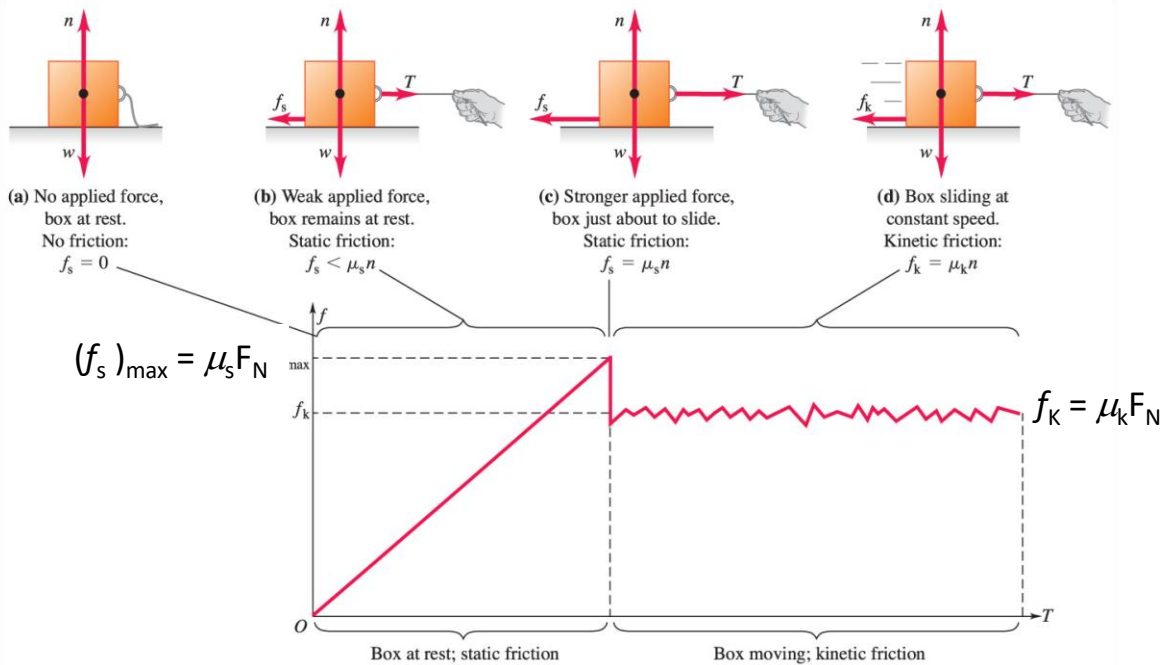


Young Freedman pg 147 Fig 5.18



[SEM image of Aluminium surface – Google Image Search](#)

Static and Kinetic Friction



(e)

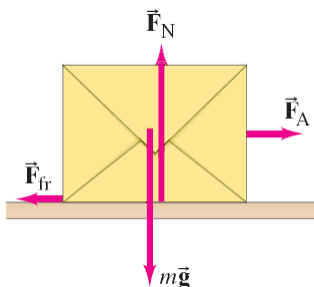
Young Freedman pg 148 Fig 5.19

1. Static friction – applicable when object is still stationary or on the verge of moving. (acts in the direction to prevent slipping)
2. Kinetic friction - applicable when object is moving. (hinders motion)

Friction: Static and Kinetic (Giancoli Example 5.1, pg 114)

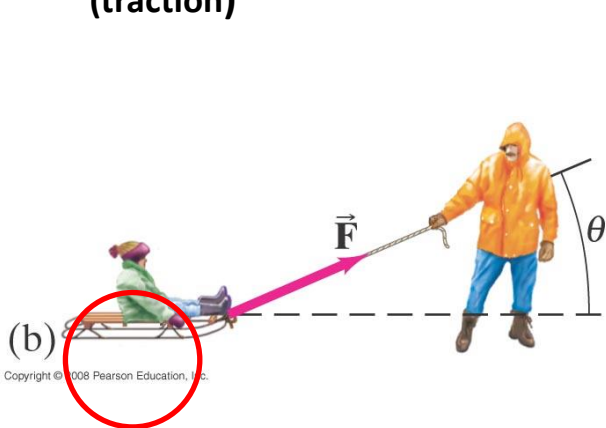
Our 10.0-kg mystery box rests on a horizontal floor. The coefficient of static friction is 0.40 and the coefficient of kinetic friction is 0.30. Determine the force of friction acting on the box if a horizontal external applied force is exerted on it of magnitude:

- (a) 10 N, (b) 38 N and (c) 40 N.

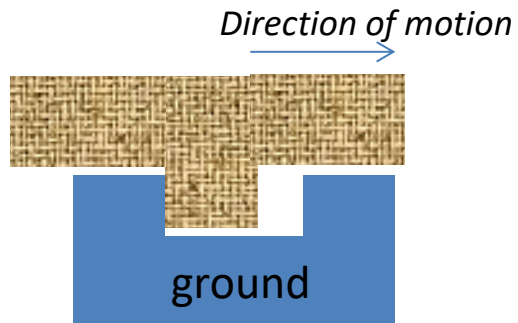
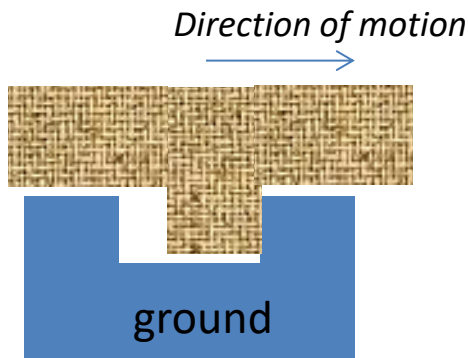


Giancoli Fig 5.5

Friction hinders motion but friction also provides grip for forward motion (traction)



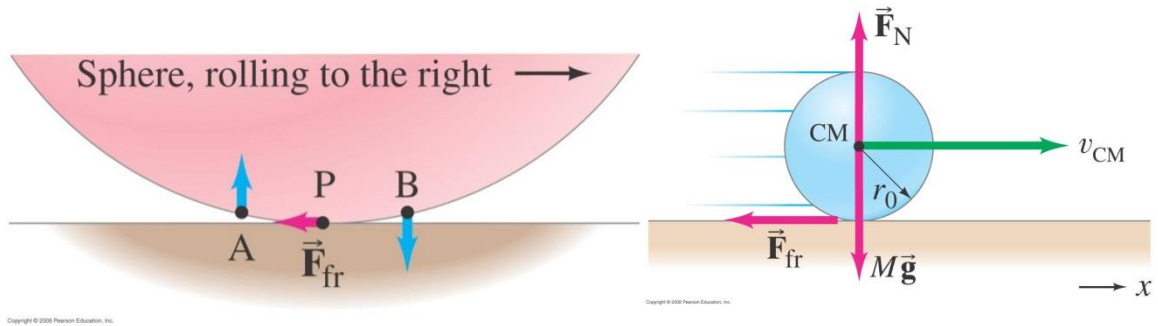
Giancoli Fig 5.6b



Not only does friction inhibit motion, it also causes wear and tear. This is particularly important in engines where specially formulated lubricants are used to reduced μ_k to as close to zero as possible. For example, modern lubricants include buckyballs – molecules consisting of 60 carbon atoms arranged in the shape of a soccer ball (acting like microscopic ball bearings).

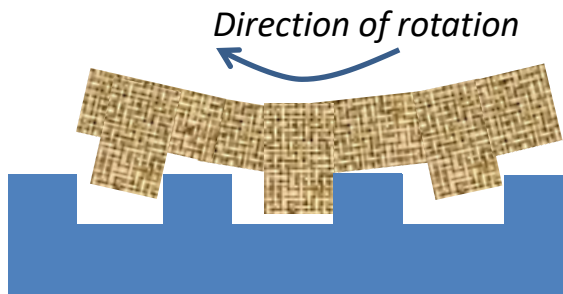
On the other hand, in Formula 1 circuits, using the right tyres with optimally high friction will allow for good traction. While most friction coefficients range between 0 to 1, top fuel race car can have friction coefficients with track surfaces of 3 or larger.

Sufficient Friction is needed for Round objects to roll: Giancoli pg 270



Giancoli pg 270 Fig 10.35, 10.38:

Without static friction, round objects would slide rather than roll. If there is not sufficient static friction, the object will roll and slide at the same time.



<https://www.youtube.com/watch?v=Rq3EsW7lywY&t=5s>